

CPE 315 Spring 2026

	Information	Comment
Instructor	John Seng Email: jseng@calpoly.edu	Lecture: MWF (10am - 21-238) Lab: MWF (11am - 14-303) Office hours: M 1p-2p (in person), T 9a-11a (Zoom), W 7p-8p (Zoom) , F 2p-3p (in person)
Course Objectives	•To understand and apply the principles of computer organization, architecture, and hardware. •To obtain a working knowledge of the MIPS instruction set architecture and CPU design. •To understand how architectural design decisions affect program performance.	
Prerequisites	CPE 203 and one of the following: CSC 225 or CPE 233	In this class we will be using assembly programming, binary number representation, and digital logic extensively. It is important that you have a good background in these areas.
Textbook	Patterson and Hennessy, <i>Computer Organization and Design (4th edition - revised)</i> , Morgan Kaufmann	
Homework	• Problems from the textbook • Typically due one week after assigned (at beginning of lecture) • 15% off for first day late, no credit after 1 day late (weekend days count) • One problem will be graded on each assignment for correctness. • Other problems graded for completeness • Readings assigned on Canvas and in class	The midterm and final exam will draw heavily from the homework material and the material covered in class. You must always show your work; the homework is the best practice for exams. You may work on homework in groups, but make sure that you try to solve the problem yourself before getting help from others in the group.
Labs	• Labs may be done with a partner. • You may change partner during quarter • Each lab assignment will specify what needs to be turned in (e.g. program, schematic, writeup) • 15% off for first day late, no credit after 1 day late (weekends count as days)	You are encouraged to study together, discuss lectures, lab concepts and computer architecture issues. List the names of anyone you talked to on lab assignments. But, DO NOT: • turn in duplicate work (even one line of code or comment) • copy work (even one line) from another student's assignment or published source (without credit) • lend another student your assignment • write part (even one line) of another student's assignment. • turn in any AI generated code Any student turning in plagiarized code (including AI generated code) will receive an F in the course.
Grading	Course grades will be based on homework, lab assignments, and tests. Grades will be curved based on final overall grades. The following will be the breakdown of the grading: 15% - Homework 25% - Labs 30% - Midterm 30% - Final exam	• Midterm/Final format to be announced • I will tell you if you are allowed to use a calculator for an exam. • Final exam on date/time in class schedule • Overall grade rubric: ○ 90+% = A ○ 80%-89% = B ○ 70%-79% = C ○ 60%-69% = D ○ 59% and below = F
Course Topics	• Number representation • MIPS instruction set • ALU, multiplication, FP • CPU structure (fetch, branch, ALU, memory) • Performance evaluation • Exceptions • Pipelining • Data dependencies, Control Hazards • Caches • Virtual Memory, Multiprocessors	• I generally do not allow laptop use during lecture as without them it allows you and other students to focus. Writing tablets and iPads are acceptable. Please see me if there is any issue regarding this.